

Lecture 6: Revenue and Profit

Maryam Alhalboni PhD.
Maryam.Alhalboni @york.ac.uk



Lecture overview



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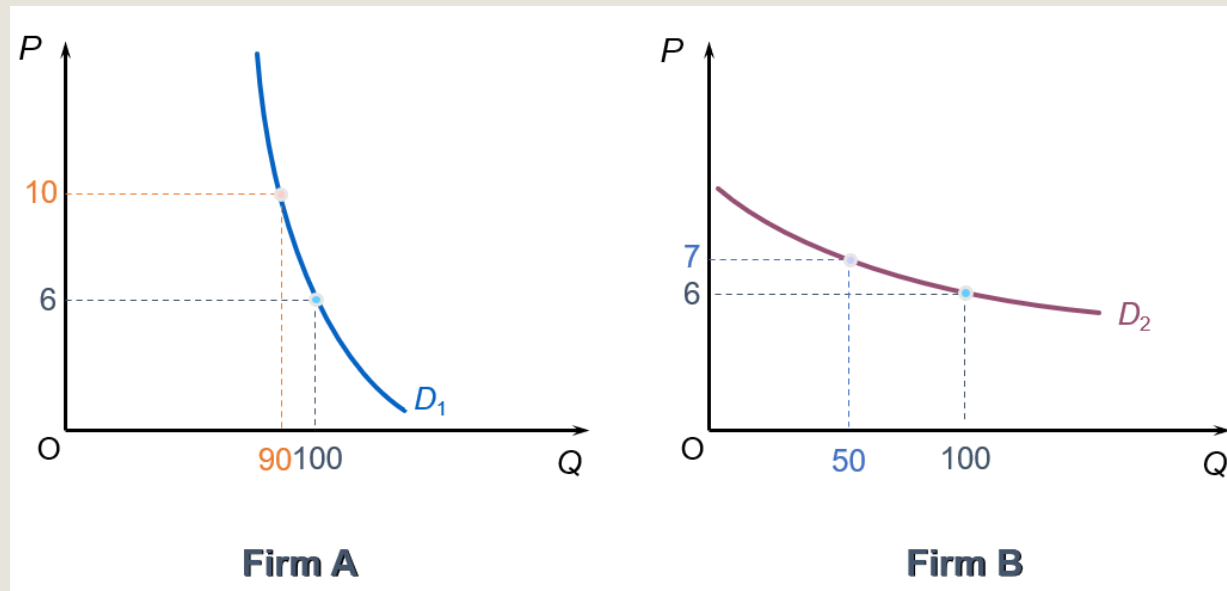
- Advertising
- Revenue
- Profit Maximization

Which statement best defines price elasticity of demand?

- A. The change in price divided by the change in income
- ☒ B. The responsiveness of quantity demanded to a change in price
- C. The responsiveness of price to a change in quantity supplied
- D. The total amount consumers spend at different prices

Which demand curve is more price elastic?

- A. Firm A
- B. Firm B
- C. Both have the same elasticity
- D. Neither can be elastic because both slope downwards



If demand is inelastic, what does this mean?

- A. A small change in price causes a large change in quantity demanded
- B. Quantity demanded does not change at all when price changes
- C. A change in price causes a relatively small change in quantity demanded
- D. Consumers buy more only when income increases

Advertising

The Role and Strategy of Advertising

The Power of Advertising

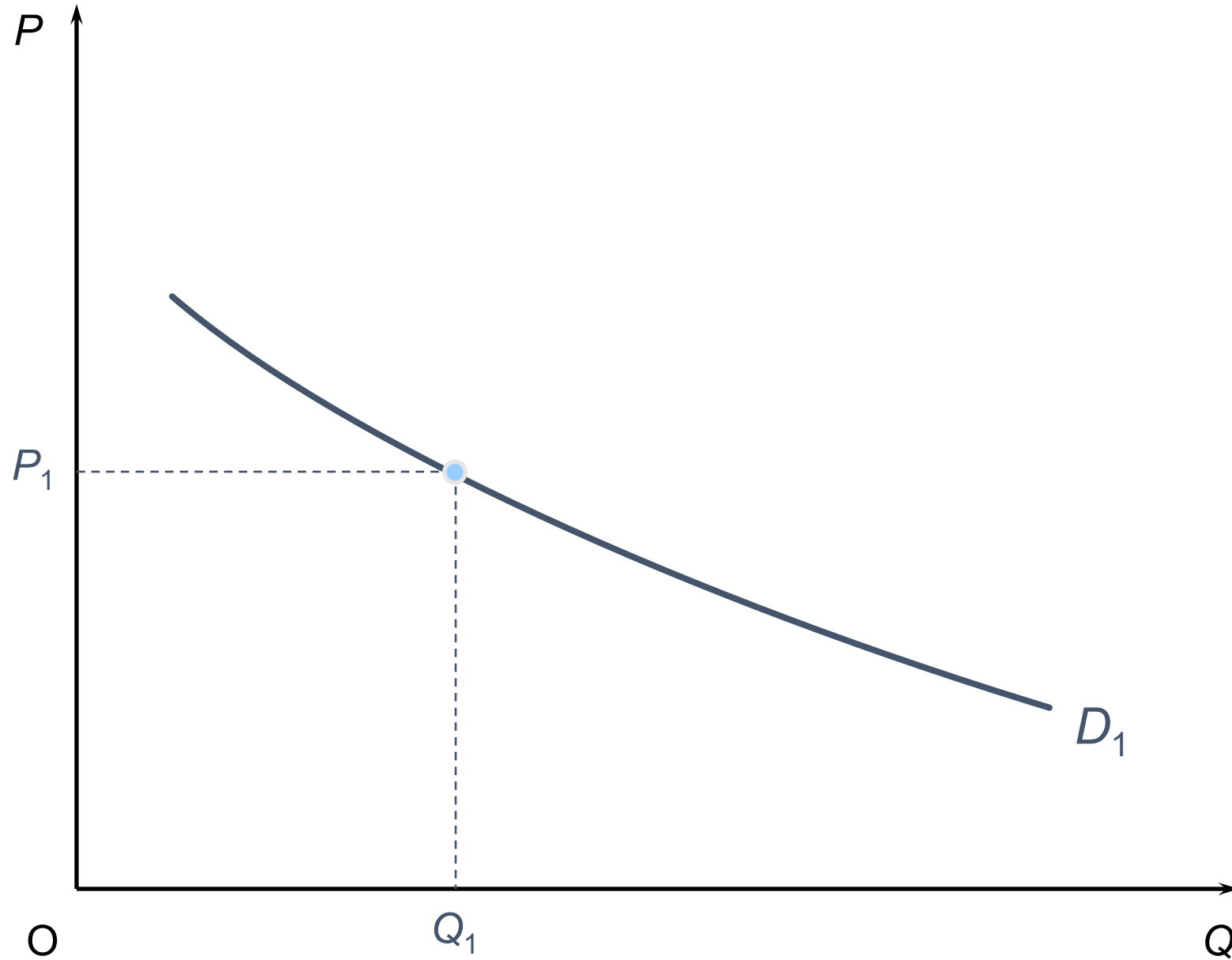
- To increase the volume of sales and drive business growth.
- Inform vs. Persuade:
 - Informing: Ensuring consumers are aware that the product exists and where they can buy it.
 - Persuading: Using psychological cues to entice the consumer, creating a desire rather than just a need.
- Competitive Differentiation: Advertising focuses on highlighting specific and unique qualities (Unique Selling Points) to make a product stand out from its rivals.
- Resource Allocation: Because of its potential for high returns, firms invest vast sums of money into advertising campaigns.

The Economic Goals of Advertising

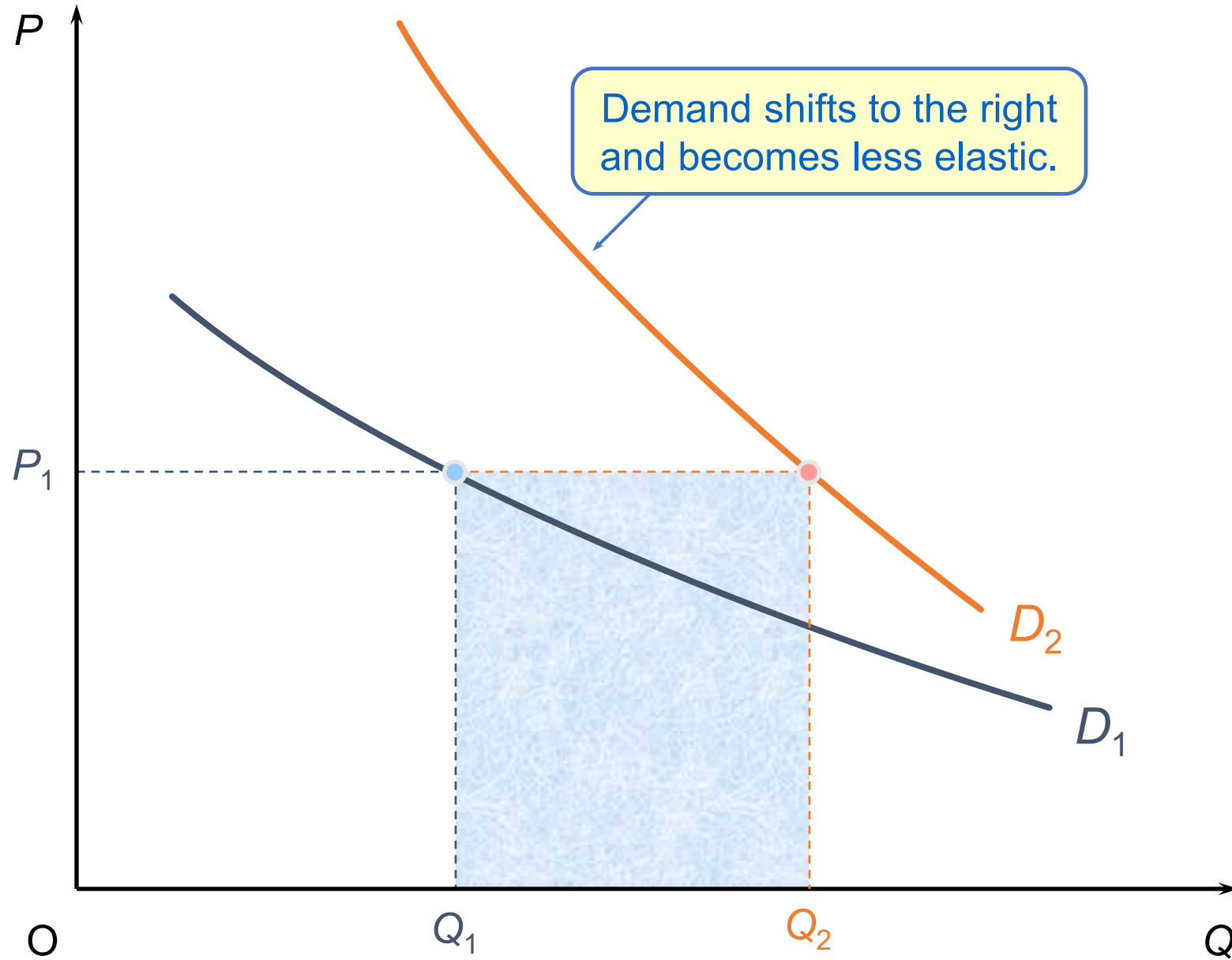
Successful advertising aims to manipulate the demand curve in two specific ways to maximize revenue:

1. Shifting the Demand Curve to the Right
 - Increase the quantity demanded at every price point.
 - By attracting new customers and increasing brand loyalty, the firm can sell more units than before without necessarily lowering the price.
2. Reducing Price Elasticity of Demand
 - Make the demand curve steeper (more inelastic).
 - By stressing unique qualities and building brand must-have status, consumers become less sensitive to price changes.
 - If demand is less elastic, the firm can raise its prices with only a small drop in the quantity sold, leading to significantly higher total revenue.

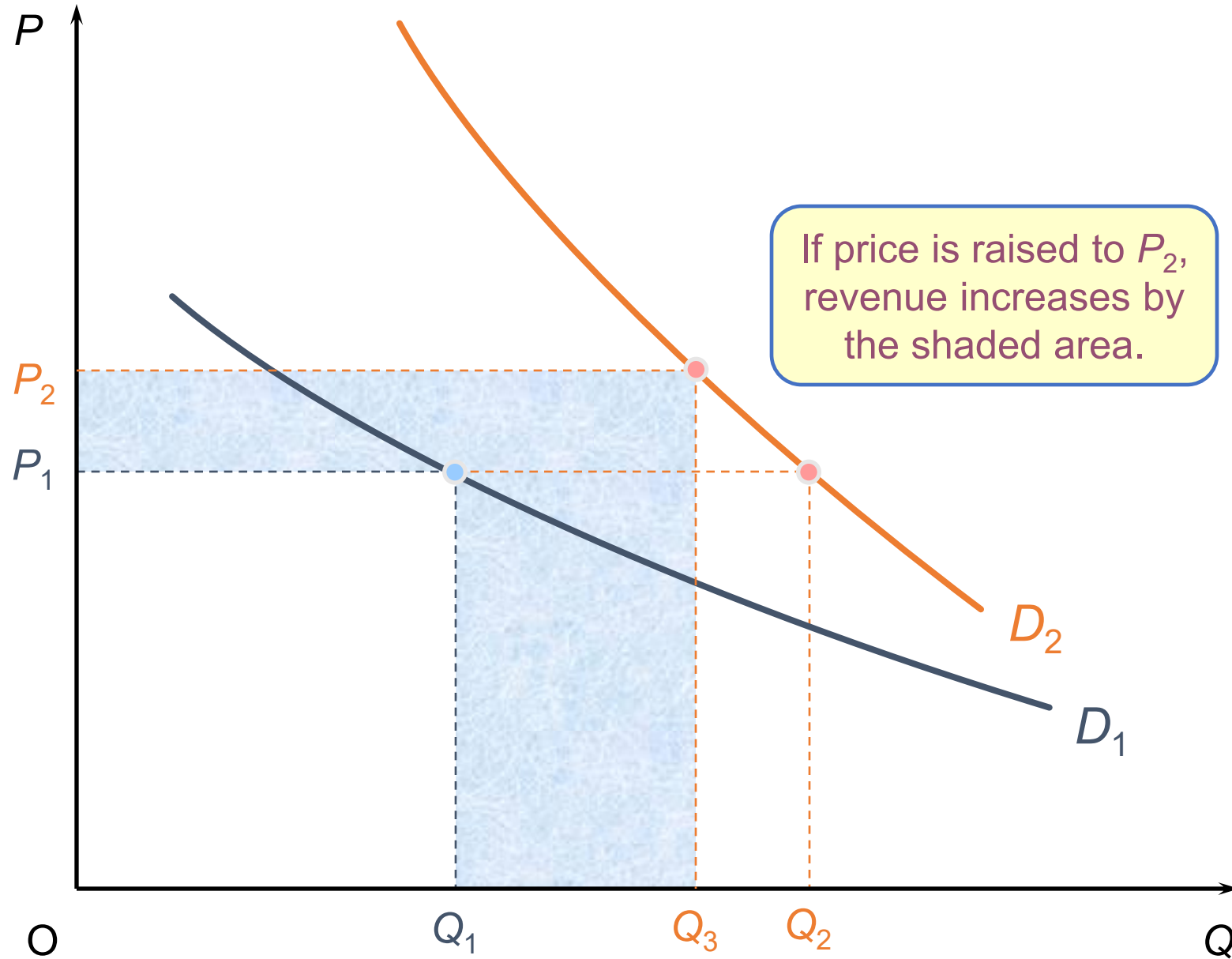
Effect of advertising on the demand curve



Effect of advertising on the demand curve



Effect of advertising on the demand curve



Analyzing the Demand Shift

- The Transition from D1 to D2:
 - D1 (Original): The firm sells quantity Q_1 at price P_1 .
 - D2 (Post-Advertising): The campaign successfully shifts the curve to the right and makes it steeper (less elastic).
- The Revenue Double-Win“
- :
 - Quantity Gain: At the original price (P_1), the firm can now sell a much larger quantity (Q_2).
 - Price Leverage: Because demand is now less elastic, the firm can raise the price to P_2 .
 - Even at a higher price, sales (Q_3) remain significantly higher than the original level (Q_1).
- The combined effect of higher prices and higher volume creates a massive increase in total revenue (the shaded area)

Mechanics of the Shift: Attracting & Persuading

How does advertising move the D curve to the right?

- Increasing Awareness: Bringing the product to the attention of new segments of the population.
- Cultivating Desire: Creating a psychological want for the product, moving it from a luxury to a perceived necessity.

Building Brand Loyalty: Reducing Elasticity

To make the curve steeper (inelastic), advertising must change consumer perception:

- Lowering Cross-Elasticity: By convincing consumers that rivals' products are inferior, the firm reduces the closeness of substitutes.
- Neutralizing the Substitution Effect: If consumers believe there are no close substitutes, they will not switch to a competitor when the firm raises its price.
- Brand Loyalty: Successful branding leads consumers to stick with the product rightly or wrongly, prioritizing brand identity over price.

The Success Metric

The more effective a campaign is, the more it achieves two simultaneous results:

- A wider shift (Volume expansion).
- A steeper slope (Pricing power)

The Profit Puzzle – Kebab Shops vs. Google

Why do some businesses barely survive while others generate billions in profit?

- Local Competition (The Kebab Shop):
 - High number of competitors.
 - Basic interiors, limited growth, and rare reinvestment.
 - Profits are often limited to Normal Profit (just enough to keep the owner in business).
- Market Dominance (Google):
 - Few real competitors.
 - Massive revenue (\$300B+) and high profit margins.
 - Significant Supernormal Profits driven by market control.

Warren Buffett's "Economic Moat"

- What is a Moat? A structural barrier that protects a company from competitors, allowing it to maintain long-term profits.
- Google's Moat:
 - Infrastructure: 2.5 million+ machines and hyperscale data centers.
 - The Barrier: To compete, a rival must match this massive hardware investment—a nearly impossible task.
- Superior technology attracts users.
 - Massive user base attracts advertisers.
 - Inelastic Demand: With few alternatives, Google gains significant pricing power.

Two Pillars of a Profitable Moat

Warren Buffett identifies two key psychological and economic factors that protect profits

1. Brand Recognition:

- Does the customer trust and recognize the name? (e.g., Coca-Cola, Apple).
- Strong brands reduce the "search cost" for consumers and build loyalty.

2. Necessity (Low Price Elasticity):

- Do customers need the product, or is it a luxury?
- Products with a low Price Elasticity of Demand (PED) allow firms to raise prices without a significant drop in sales volume

The Payoff of Protecting Profits

- Type equation here. Investment Philosophy: Buy businesses with high moats and hold them forever.
- Real-World Returns (as of 2024):
 - American Express: \$1.3B investment → \$34.5B
 - Coca-Cola: \$1.3B investment → \$24.5B
 - Apple: \$31B investment → \$135.4B
- Markets aren't just about selling a product; they are about managing competition.

Revenue and Profit

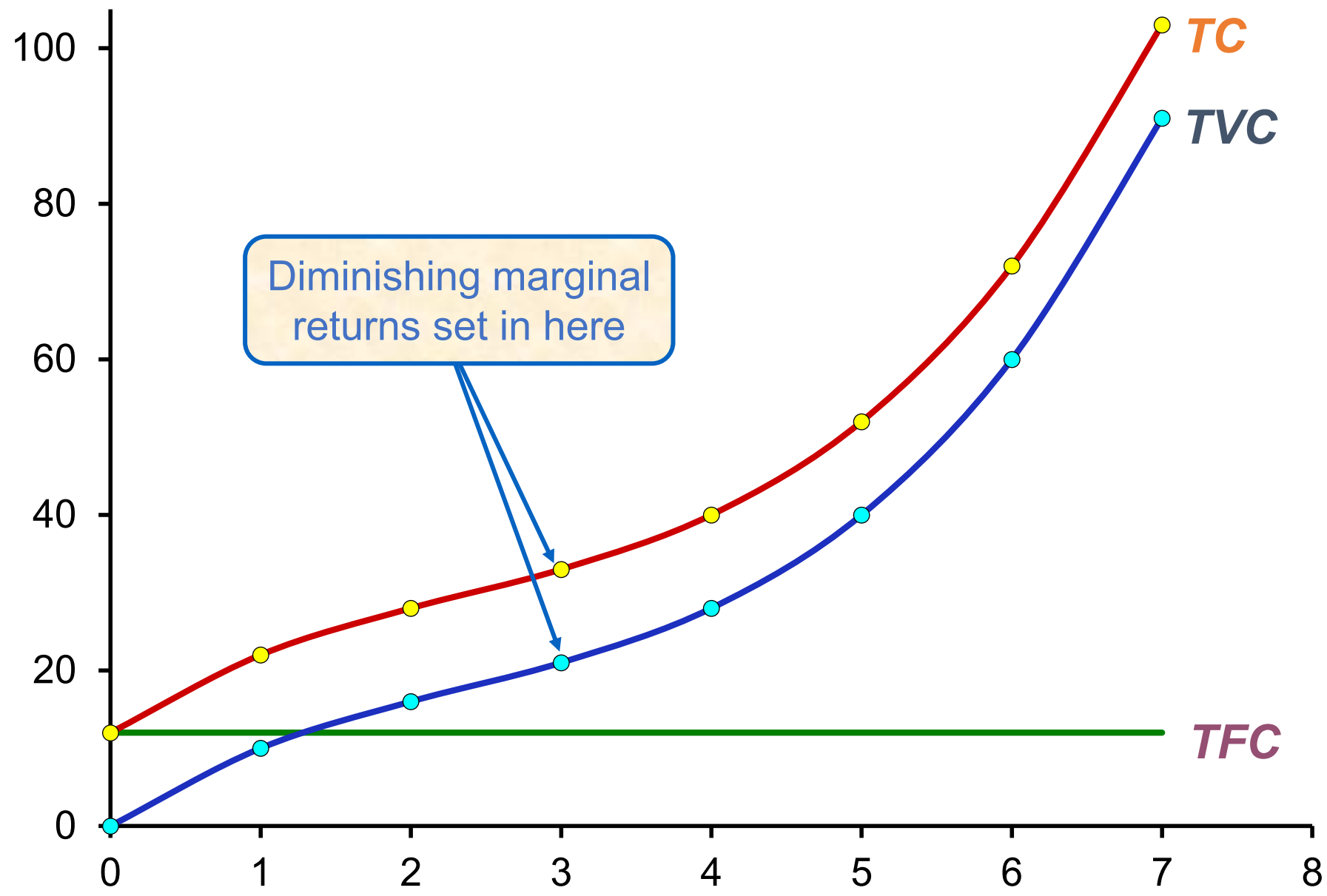
Which description best matches the typical shape of a short-run total cost (STC) curve?



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- A. It starts at zero and rises at a constant rate
- ☒ B. It starts above zero (fixed cost) and rises first at a decreasing rate, then at an increasing rate
- C. It starts below zero and rises slowly throughout
- D. It is horizontal because costs do not change in the short run

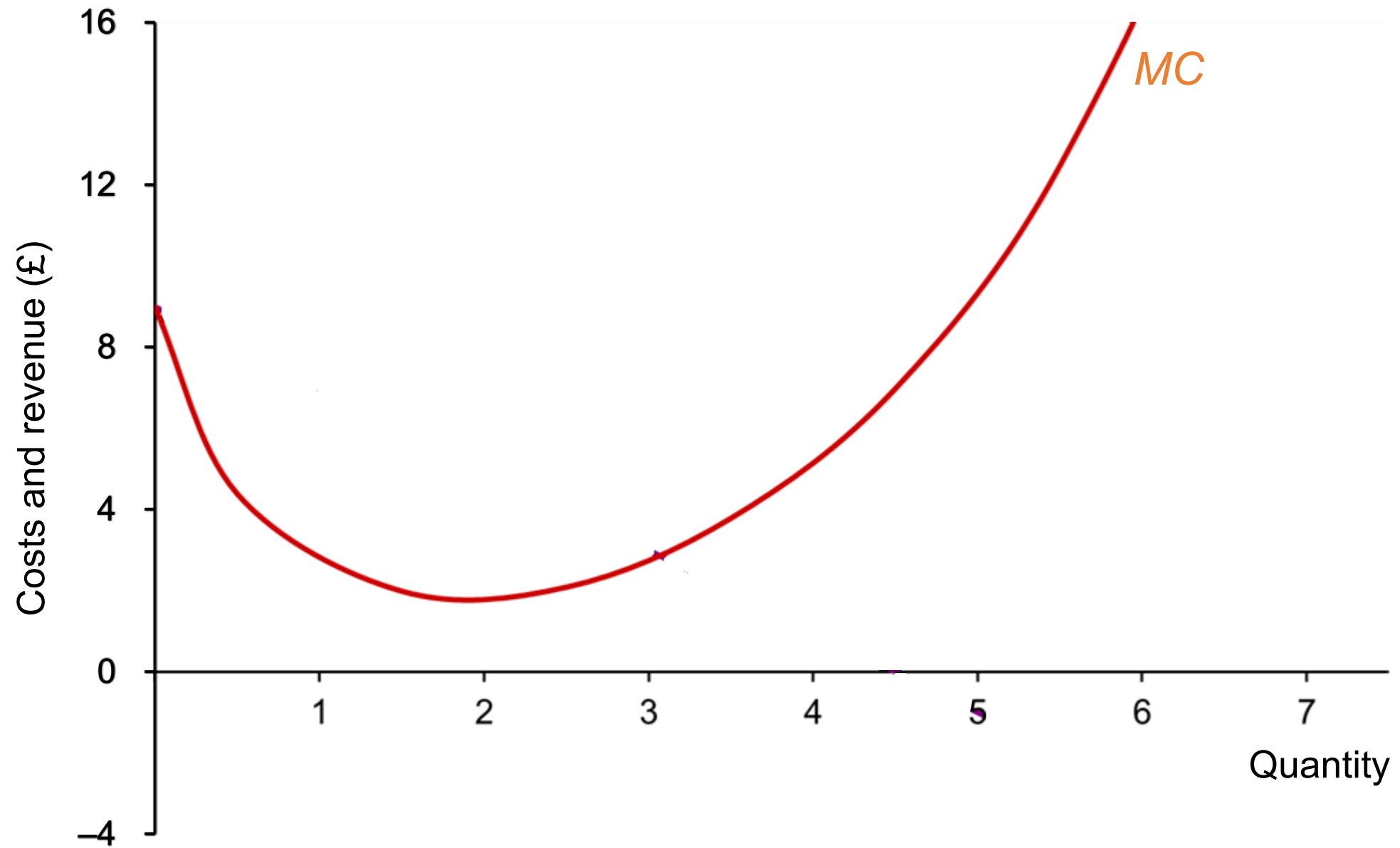
Total costs for firm X



Output (Q)	TFC (£)	TVC (£)	TC (£)
0	12	0	12
1	12	10	22
2	12	16	28
3	12	21	33
4	12	28	40
5	12	40	52
6	12	60	72
7	12	91	103

Which statement best describes the marginal cost (MC) curve in the short run?

- A. MC is always horizontal because marginal cost is constant
- ☒ B. MC is typically U-shaped: it falls at first, then rises due to diminishing returns
- C. MC always slopes downward because firms become more efficient
- D. MC is the same as average cost at every output level

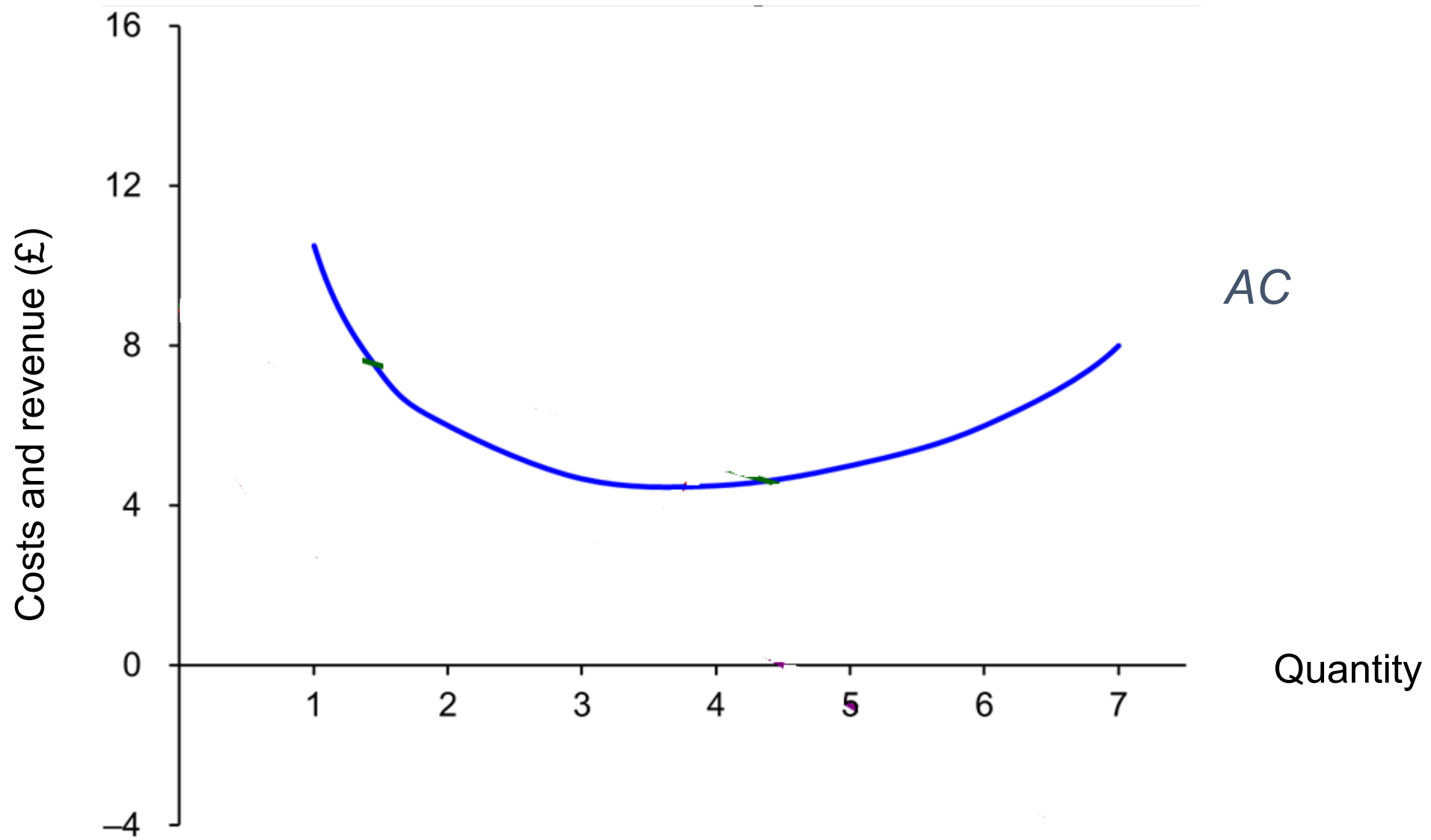


Which statement best describes the short-run average cost (SRAC) curve?



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- ☒ A. SRAC is typically U-shaped because average cost falls at first (spreading fixed costs) and later rises due to diminishing returns.
- ☐ B. SRAC is a horizontal line because average cost is constant in the short run.
- ☐ C. SRAC always rises as output increases because fixed costs increase in the short run.
- ☐ D. SRAC always falls as output increases because variable costs never rise.



The Profit Equation

- To identify the specific price and output level that maximizes total profit.
- Profit: The difference between what a firm earns and what it spends.

$$\pi = TR - TC$$

- Revenue Concepts: Just as with costs, economists analyze revenue through three lenses: Total, Average, and Marginal.

The Three Revenue Pillars

- Total Revenue (TR): The total earnings from sales over a period.

$$TR = P \times Q$$

- Example: Selling 1,000 units at £5 each results in $TR = £5,000$.

- Average Revenue (AR): The amount earned per unit sold.

$$AR = TR / Q$$

- Under most conditions, $AR = \text{Price}$. (The only exception is price discrimination, where AR is the weighted average price).

The Three Revenue Pillars

- Marginal Revenue (MR): The additional revenue generated by selling one extra unit.

$$MR = \Delta TR / \Delta Q$$

- Example: If selling 20 extra units increases TR by £100, the MR is £5 per unit.

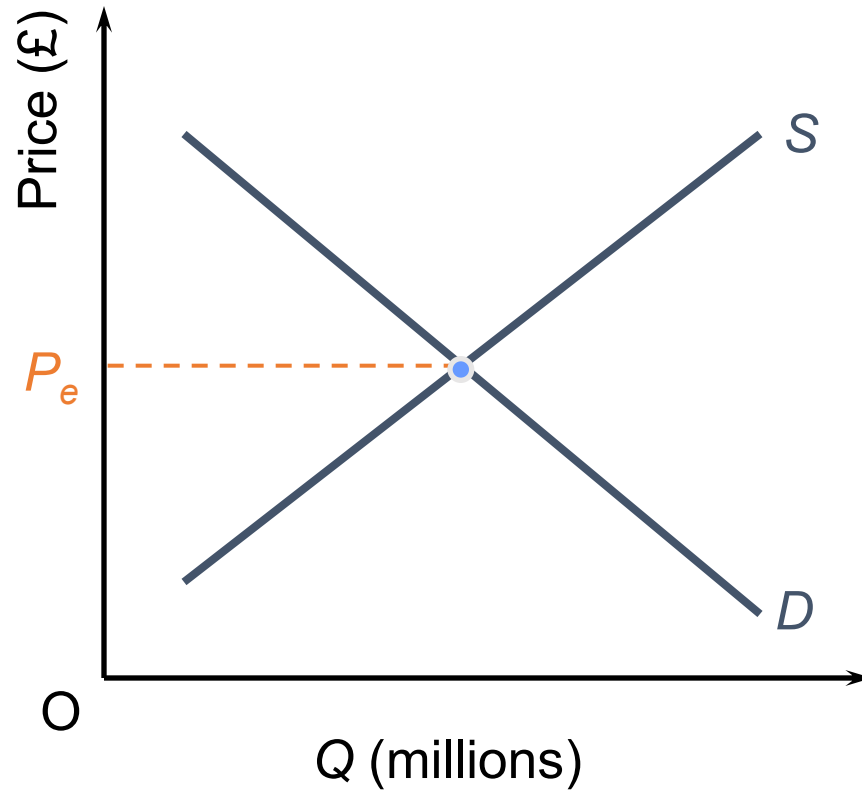
Revenue Curves and Market Power

The shape of TR, AR, and MR curves is not fixed; it changes based on the firm's ability to influence market prices.

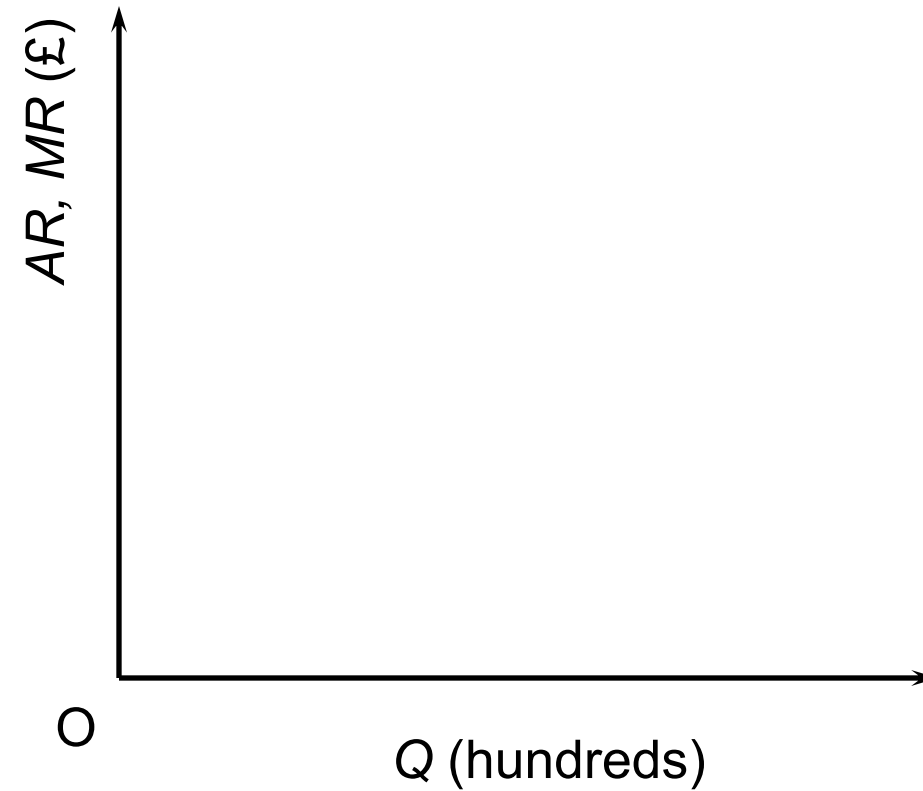
Two Distinct Market Scenarios

1. The Price Taker (Small Firm):
 - The firm is too small to affect the market price.
 - It must sell all its output at the prevailing market price.
 - The AR and MR curves will be horizontal.

Deriving a firm's *AR* and *MR*: price-taking firm

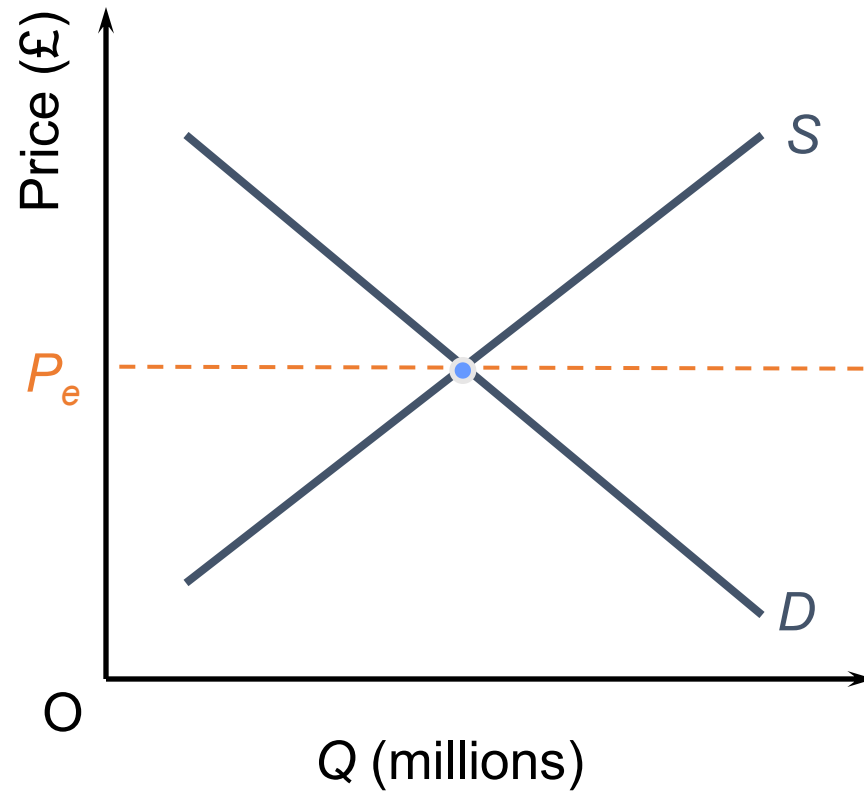


(a) The market

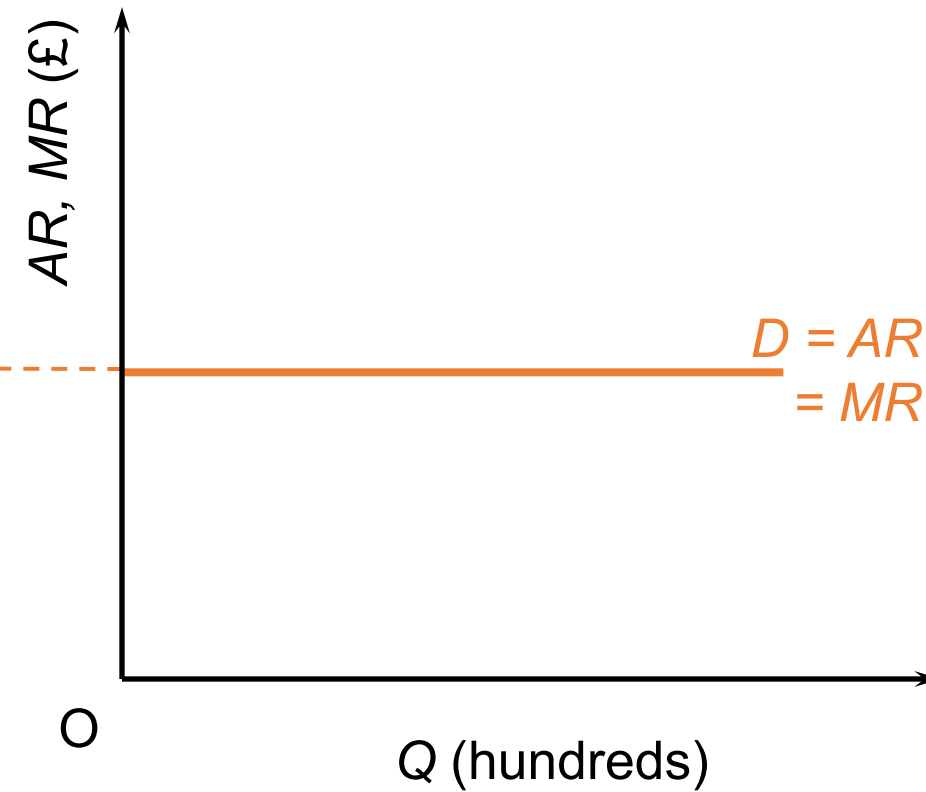


(b) The firm

Deriving a firm's *AR* and *MR*: price-taking firm

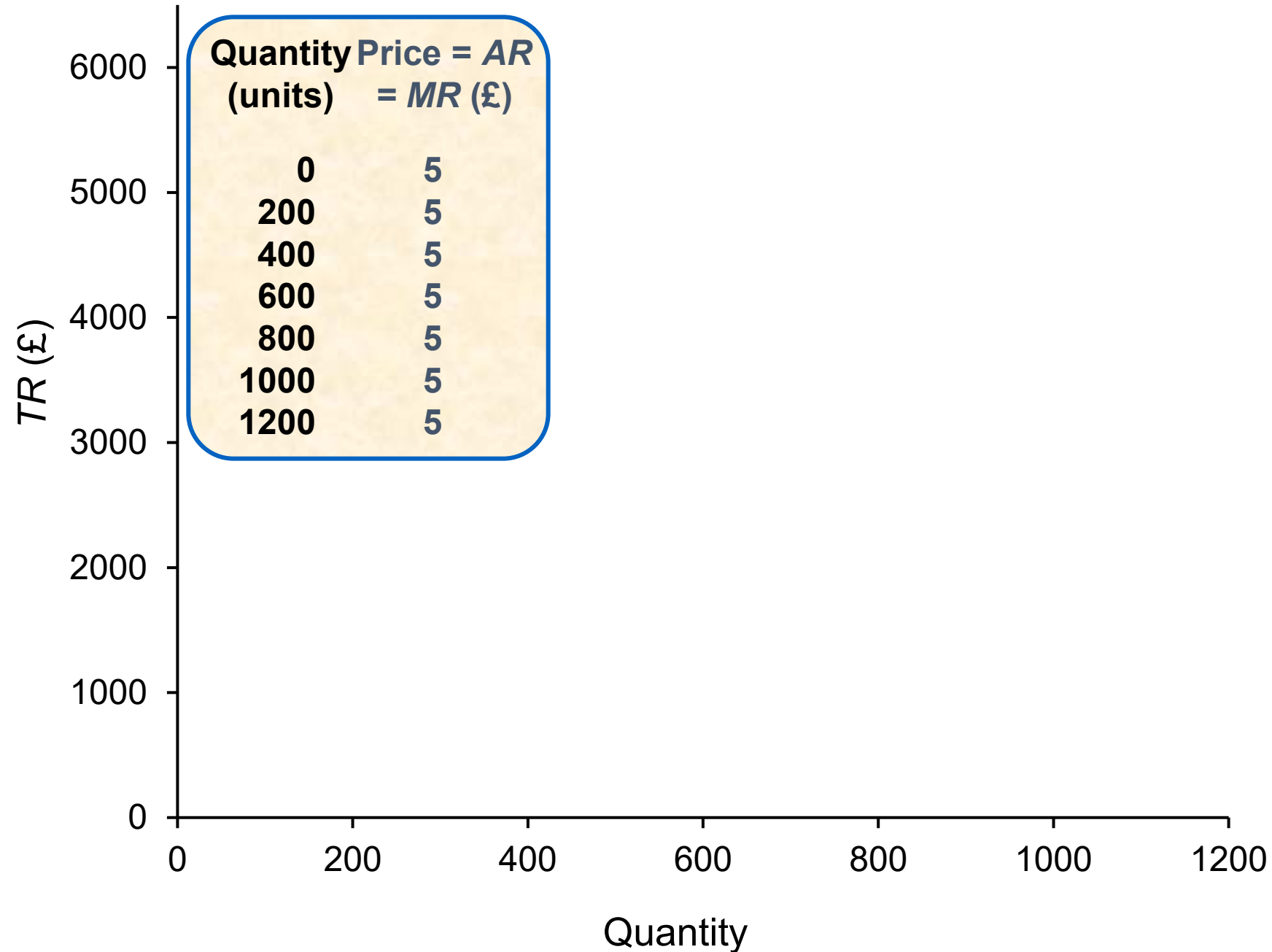


(a) The market

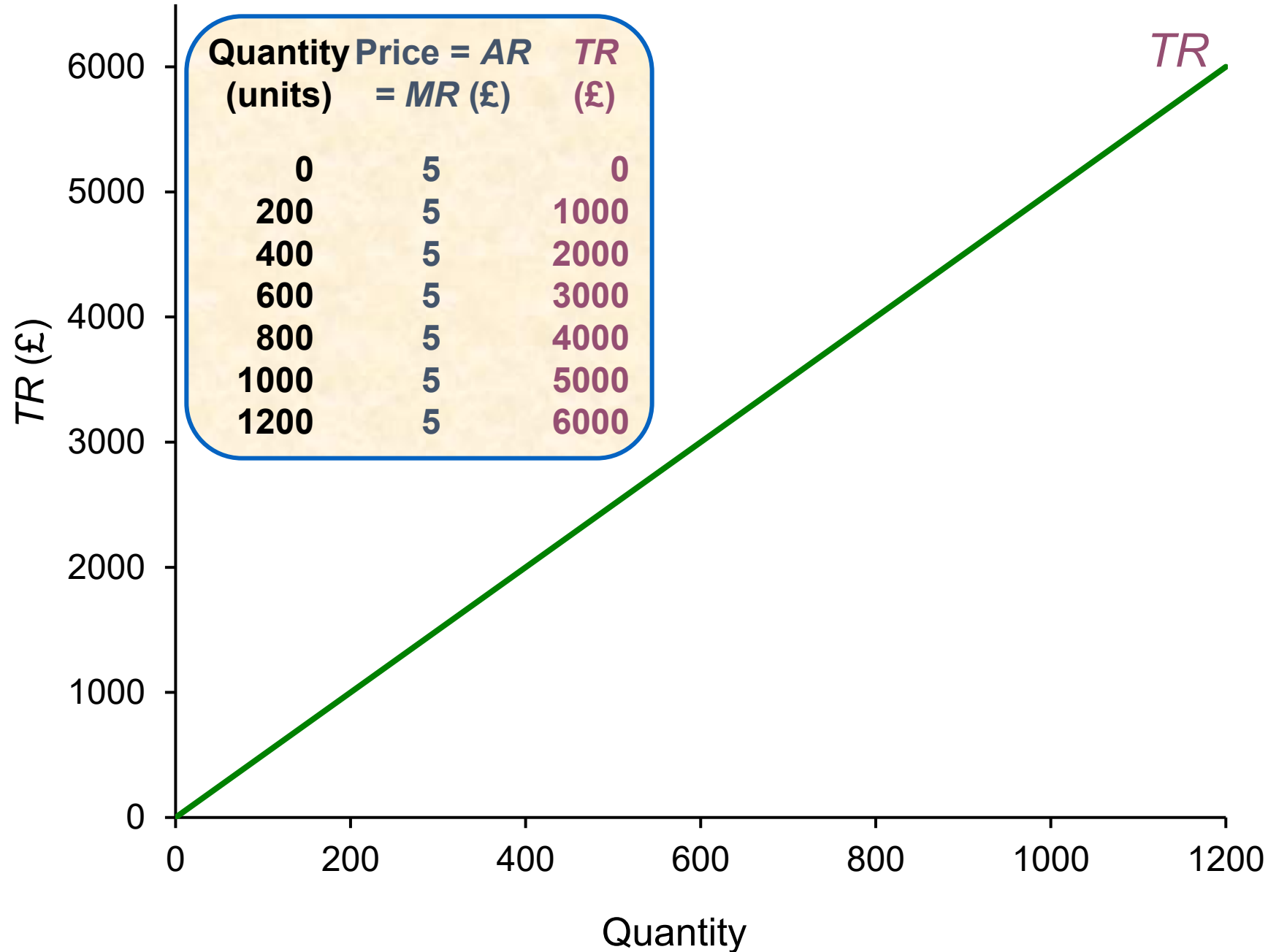


(b) The firm

Total revenue for a price-taking firm



Total revenue for a price-taking firm



Q A price-taker is a firm which has:

- A. control over its prices.
- B. its prices fixed by a regulatory body.
- C. a strong incentive to advertise.
- ☒ D. no control over its price.

Revenue Curves and Market Power

Two Distinct Market Scenarios

1. The Price Taker (Small Firm):
2. The Price Maker (Firm with Market Power):
 - The firm has the choice to set its price.
 - To sell more, it must usually lower its price.
 - AR and MR curves will be downward-sloping.

Revenue Curves for the Price Maker

Unlike price takers, these firms have market power. They face a downward-sloping demand curve.

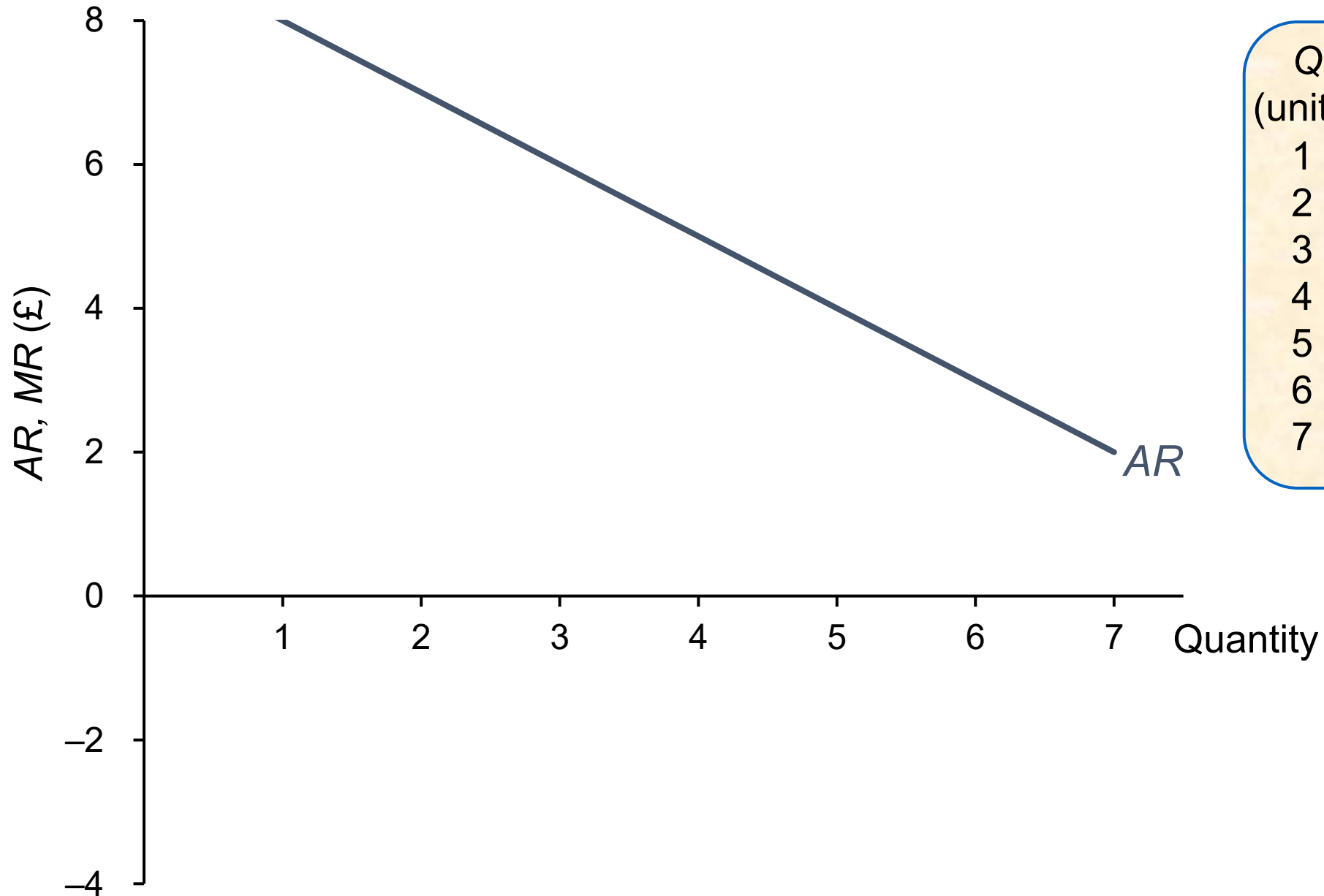
The Trade-off:

- To sell a higher quantity, the firm must lower its price.
- Alternatively, it can raise the price, but it must accept a lower quantity sold.

Average Revenue (AR):

- Since $AR = Price$, the AR curve is identical to the demand curve.
- As price falls to increase sales, AR must also fall.

AR and MR curves for a firm facing a downward-sloping demand curve



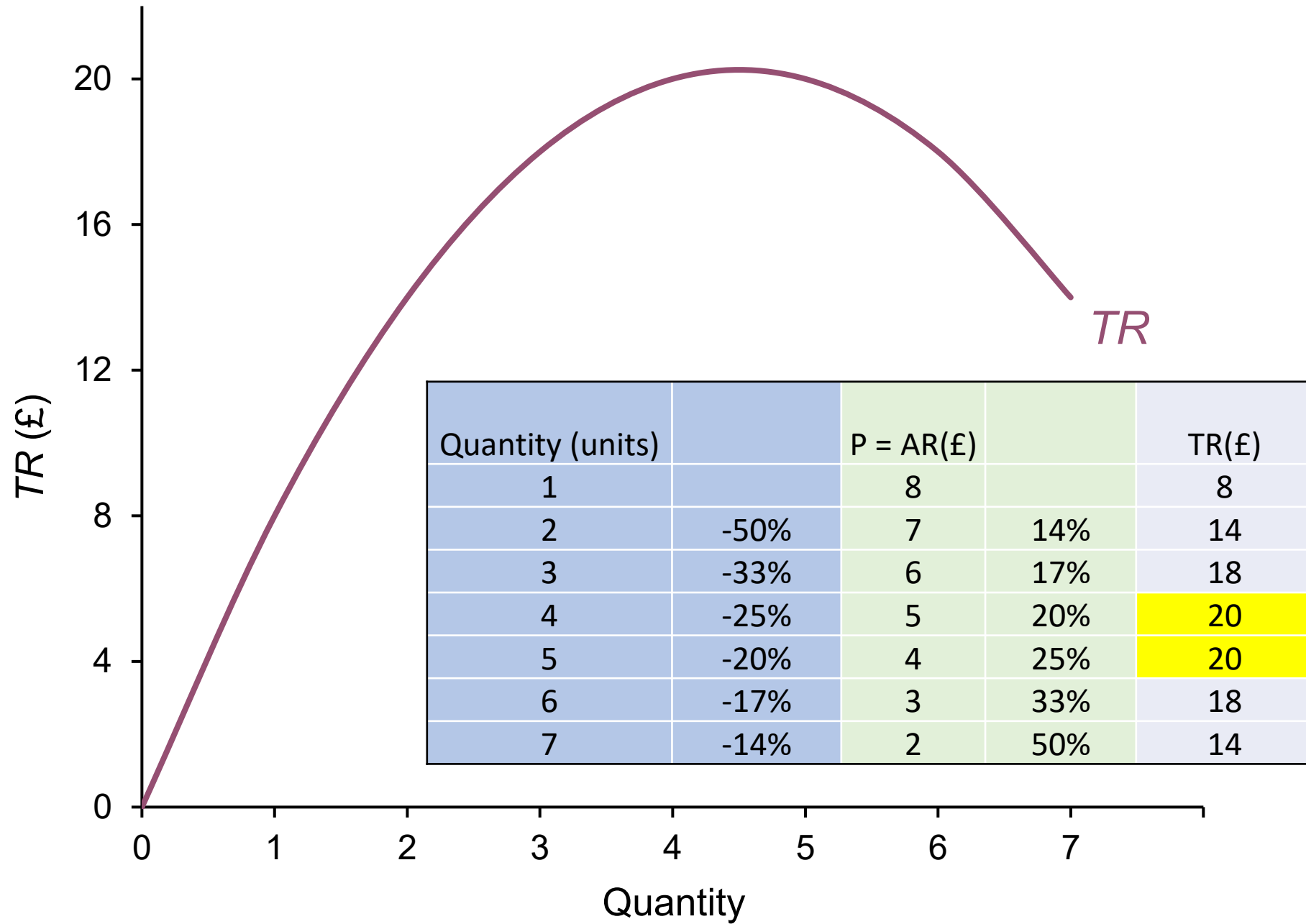
Q (units)	P = AR (£)	TR (£)	MR (£)
1	8	8	
2	7	14	6
3	6	18	4
4	5	20	2
5	4	20	0
6	3	18	-2
7	2	14	-4

Total Revenue (TR) Dynamics

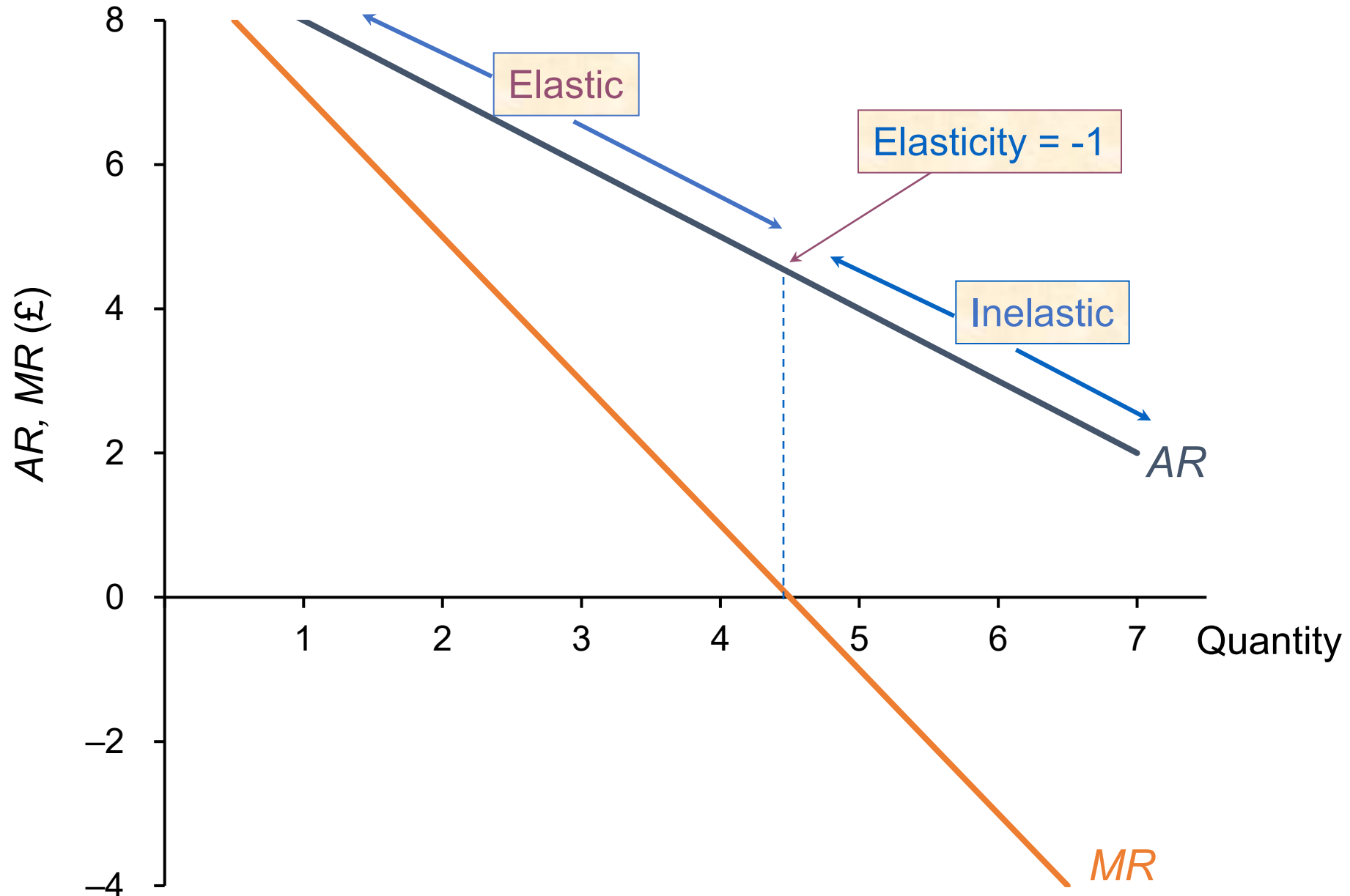
The Non-Linear TR Curve

- For a price maker, the TR curve is an inverted U-shape (it rises, peaks, and then falls).
- lowering the price increases sales enough to raise total revenue.
- TR reaches its maximum at the point where Marginal Revenue (MR) equals zero.
- Eventually, further price cuts reduce revenue more than the extra sales gain, causing TR to fall.

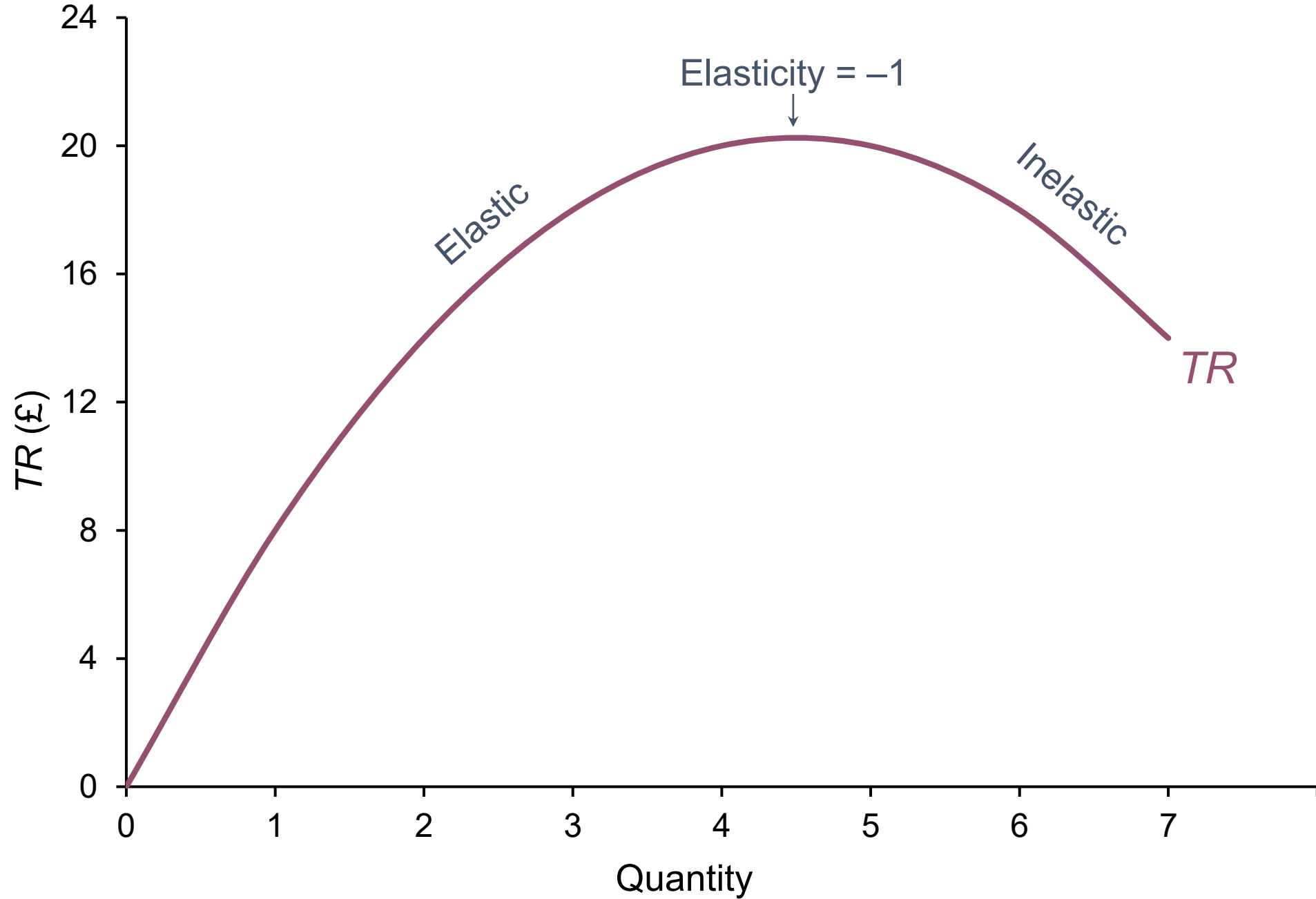
TR curve for a firm facing a downward-sloping D curve



***AR* and *MR* curves for a firm facing a downward-sloping demand curve**



***TR* curve for a firm facing a downward-sloping *D* curve**



Profit Maximisation

Profit Maximization in the Short Run

- We can now combine cost and revenue data to find the profit-maximizing output level.
- Two Methods of Calculation:
 1. Total Curves Method: Compare TR and TC. Profit is maximized where the vertical gap between TR and TC is at its widest.
 2. Marginal/Average Method: Find where $MC = MR$. This is the standard rule used to identify the exact point of maximum profit.

Profit Maximization in the Short Run

Key Assumptions for This Analysis:

- Short Run Focus: At least one factor of production is fixed (Fixed Costs exist).
- Firm Type: We are focusing on a Price Maker (downward-sloping demand).

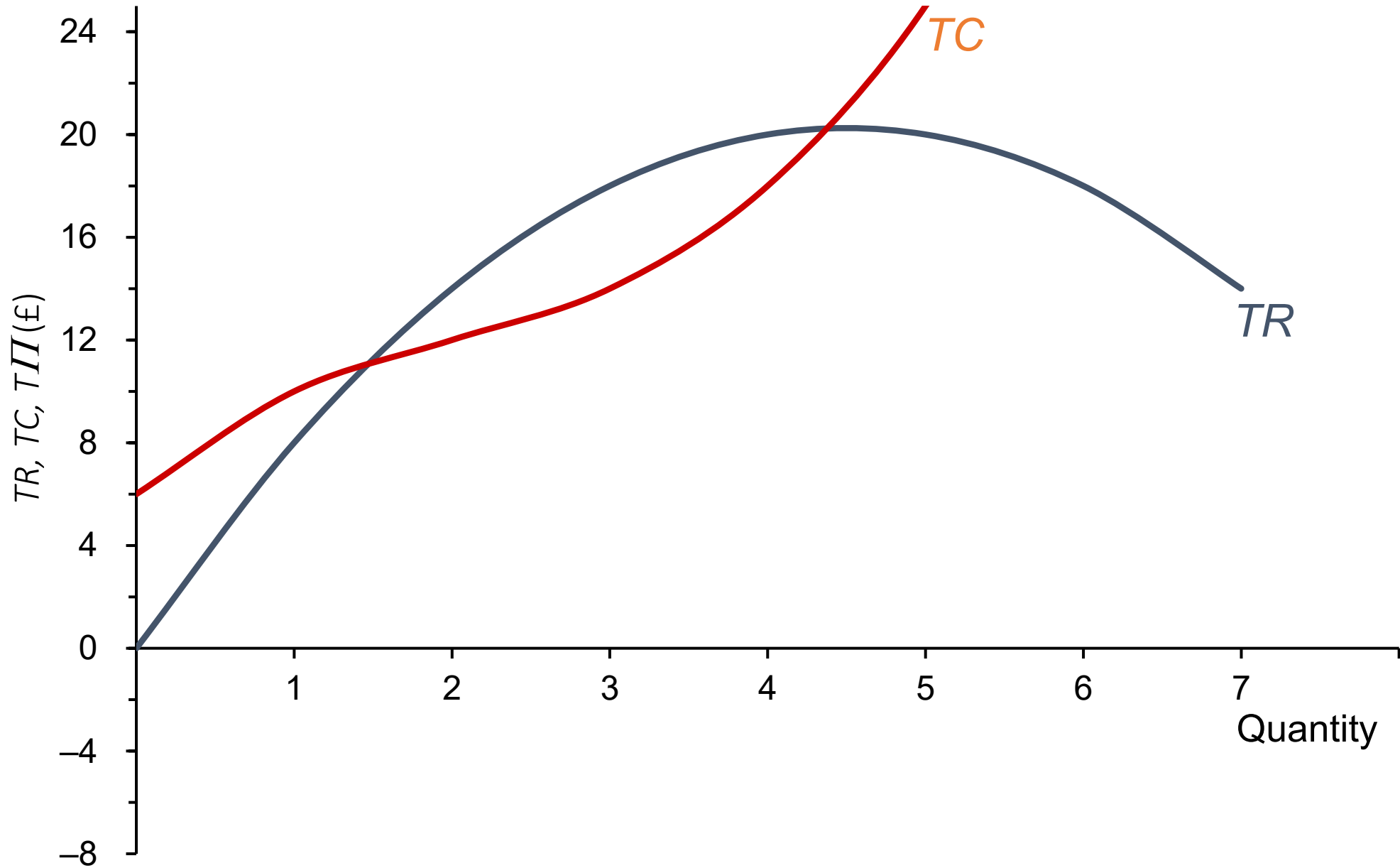
Revenue, cost and profit for Firm X

Q (units)	$P = AR$ (£)	TR (£)	MR (£)	TC (£)	AC (£)	MC (£)	TΠ (£)	AΠ (£)
0	9	0		6	—		−6	—
			8			4		
1	8	8		10	10		−2	−2
			6			2		
2	7	14		12	6		2	1
			4			2		
3	6	18		14	$4\frac{2}{3}$		4	$1\frac{1}{3}$
			2			4		
4	5	20		18	$4\frac{1}{2}$		2	$\frac{1}{2}$
			0			7		
5	4	20		25	5		−5	−1
			−2			11		
6	3	18		36	6		−18	−3
			−4			20		
7	2	14		56	8		−42	−6

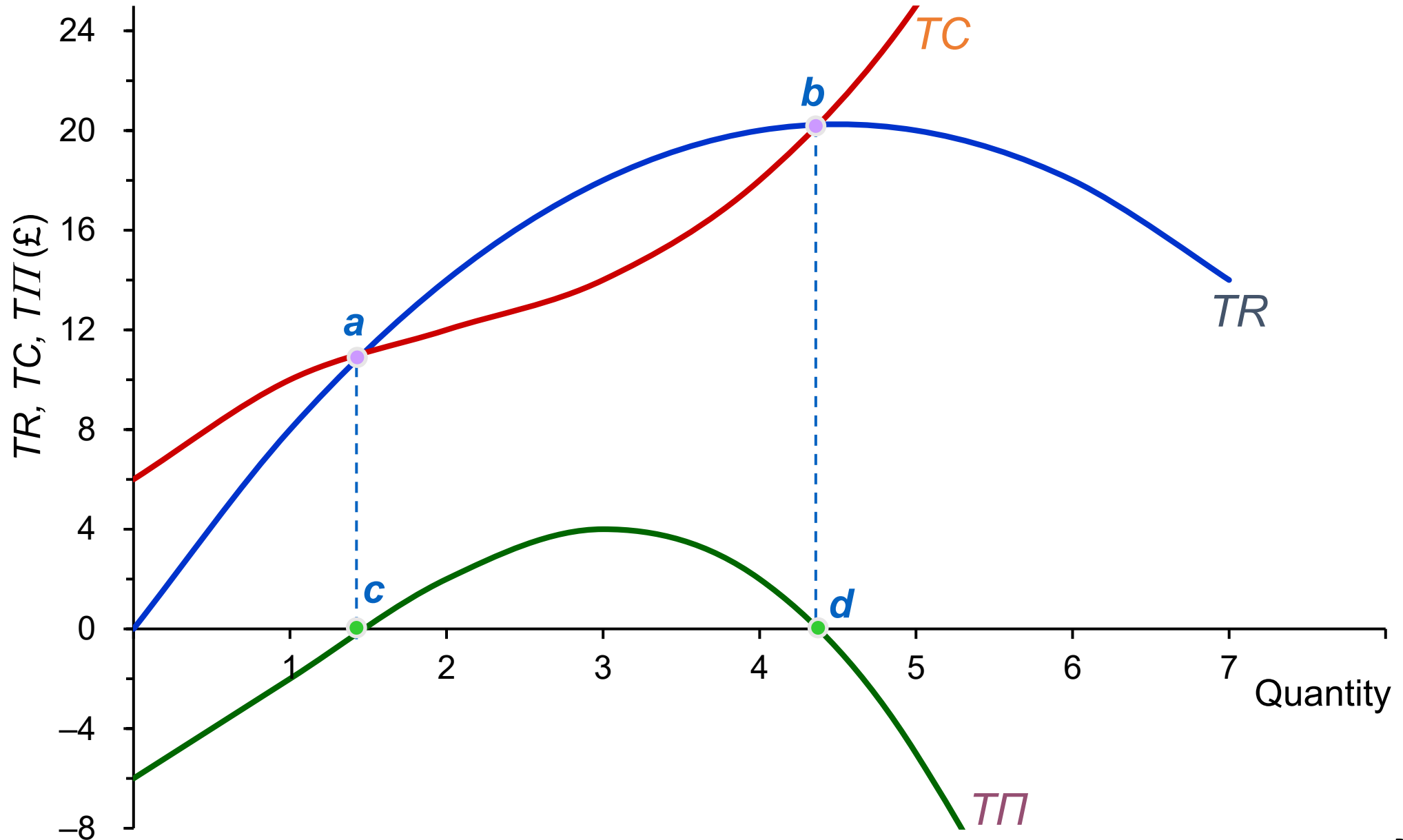
Profit Maximization (Total Curves Method)

- Total profit is the vertical distance between Total Revenue (TR) and Total Cost (TC). Maximum Profit Point: Occurs at the output level where the gap between TR and TC is at its widest.
- At 3 units of output: $TR = £18$ and $TC = £14$.
- Total Profit: $£18 - £14 = £4$.
- Below 3 units, the gap is smaller (or negative if $TC > TR$).
- Above 3 units, the gap begins to shrink as diminishing returns increase costs faster than revenue grows.

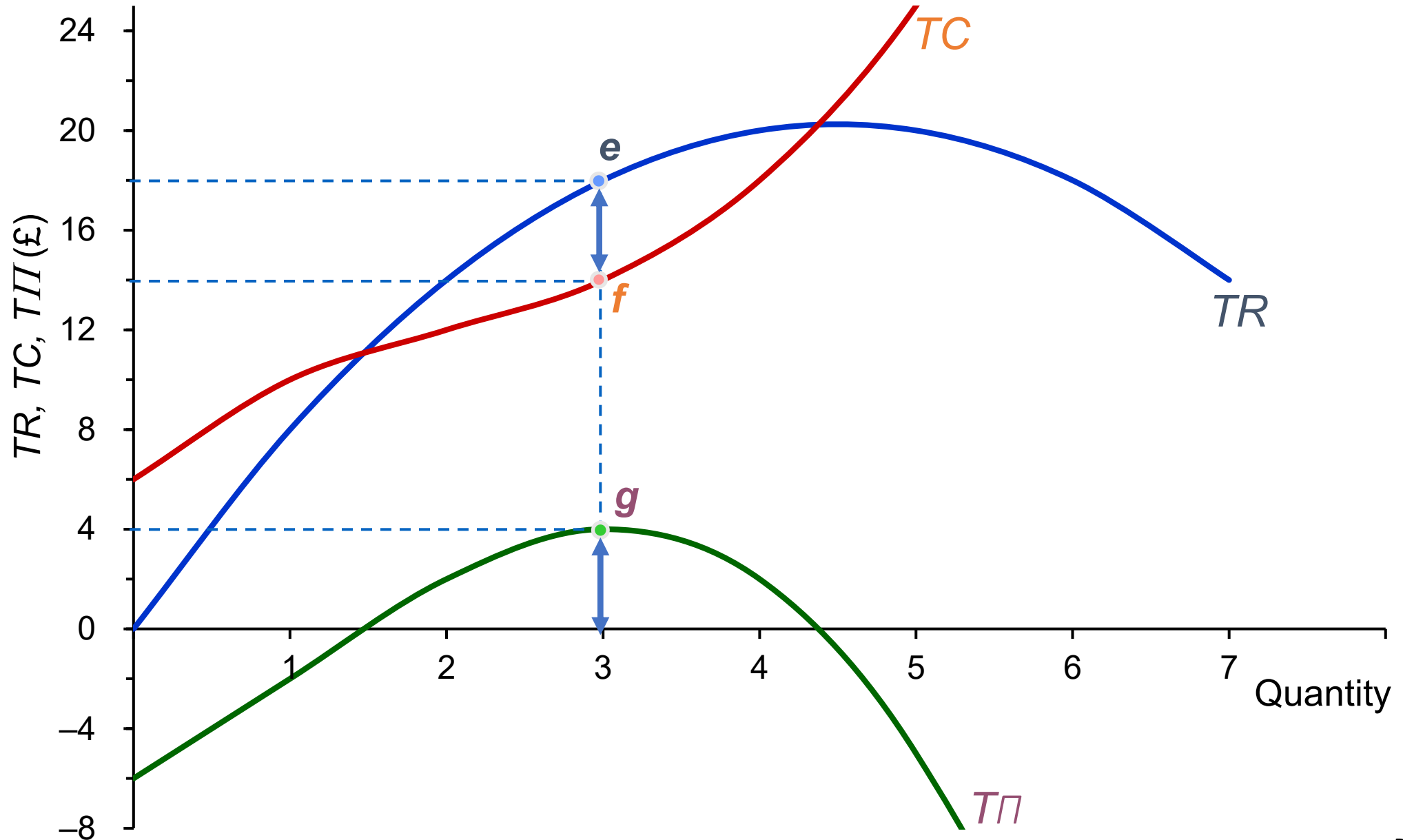
Finding maximum profit using total curves



Finding maximum profit using total curves



Finding maximum profit using total curves



Profit Maximization (Marginal Method)

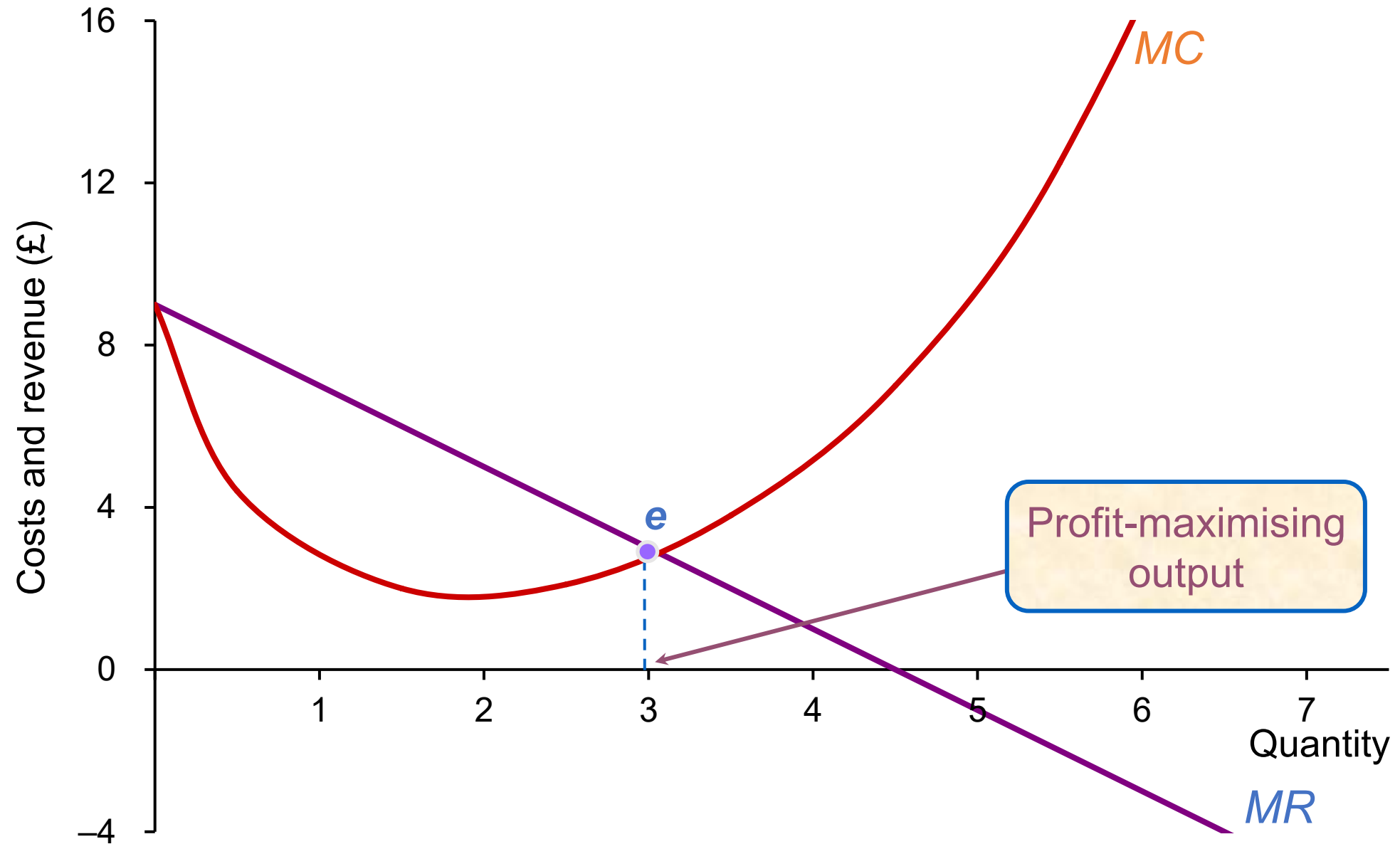
The Two-Stage Process

- Stage 1: Find the Profit-Maximizing Output using Marginal Revenue (MR) and Marginal Cost (MC).
- Stage 2: Calculate the Total Profit at that output using Average Revenue (AR) and Average Cost (AC).

Stage 1: The Golden Rule ($MC = MR$)

- The Rule: Profits are maximized at the exact point where $MC = MR$.
- Why this works:
 - If $MR > MC$: Every extra unit sold adds more to revenue than to cost. The firm should increase production to grow profit.
 - If $MC > MR$: The last unit produced cost more to make than it earned. The firm should decrease production to grow profit.
 - Equilibrium: At $MC = MR$, there are no further gains to be made by changing output.

Finding the profit-maximising output using marginal curves



Stage 1: The Golden Rule ($MC = MR$)

- The Rule: Profits are maximized at the exact point where $MC = MR$.
- Why this works:
 - If $MR > MC$: Every extra unit sold adds more to revenue than to cost. The firm should increase production to grow profit.
 - If $MC > MR$: The last unit produced cost more to make than it earned. The firm should decrease production to grow profit.
 - Equilibrium: At $MC = MR$, there are no further gains to be made by changing output.

Measuring Supernormal Profit

Stage 2: Calculating Total Profit

Once the output is set (where $MC=MR$), we use the Average curves:

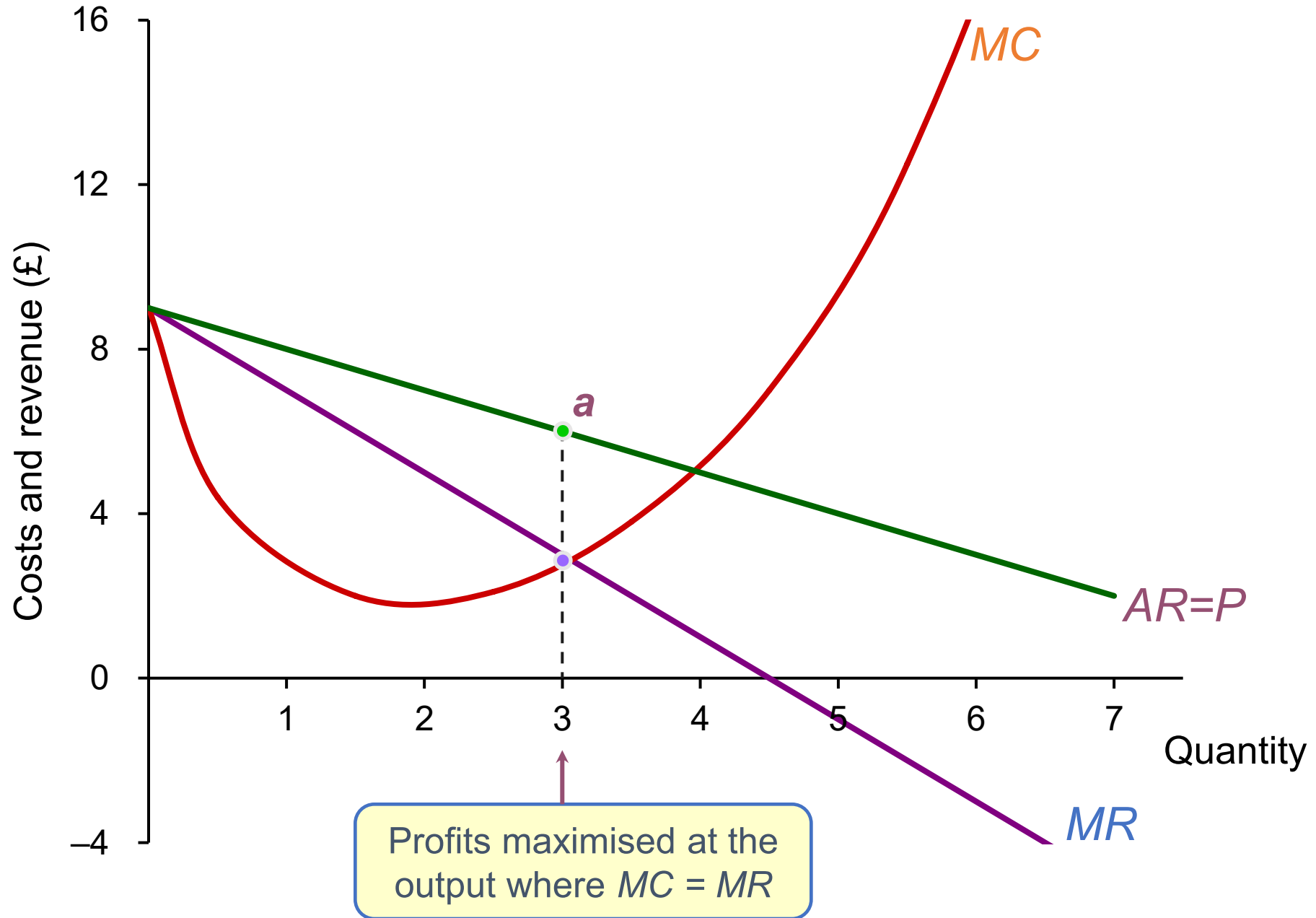
- Average Profit ($A\pi$): The difference between what you get per unit (AR) and what it costs per unit (AC).

$$A\pi = AR - AC$$

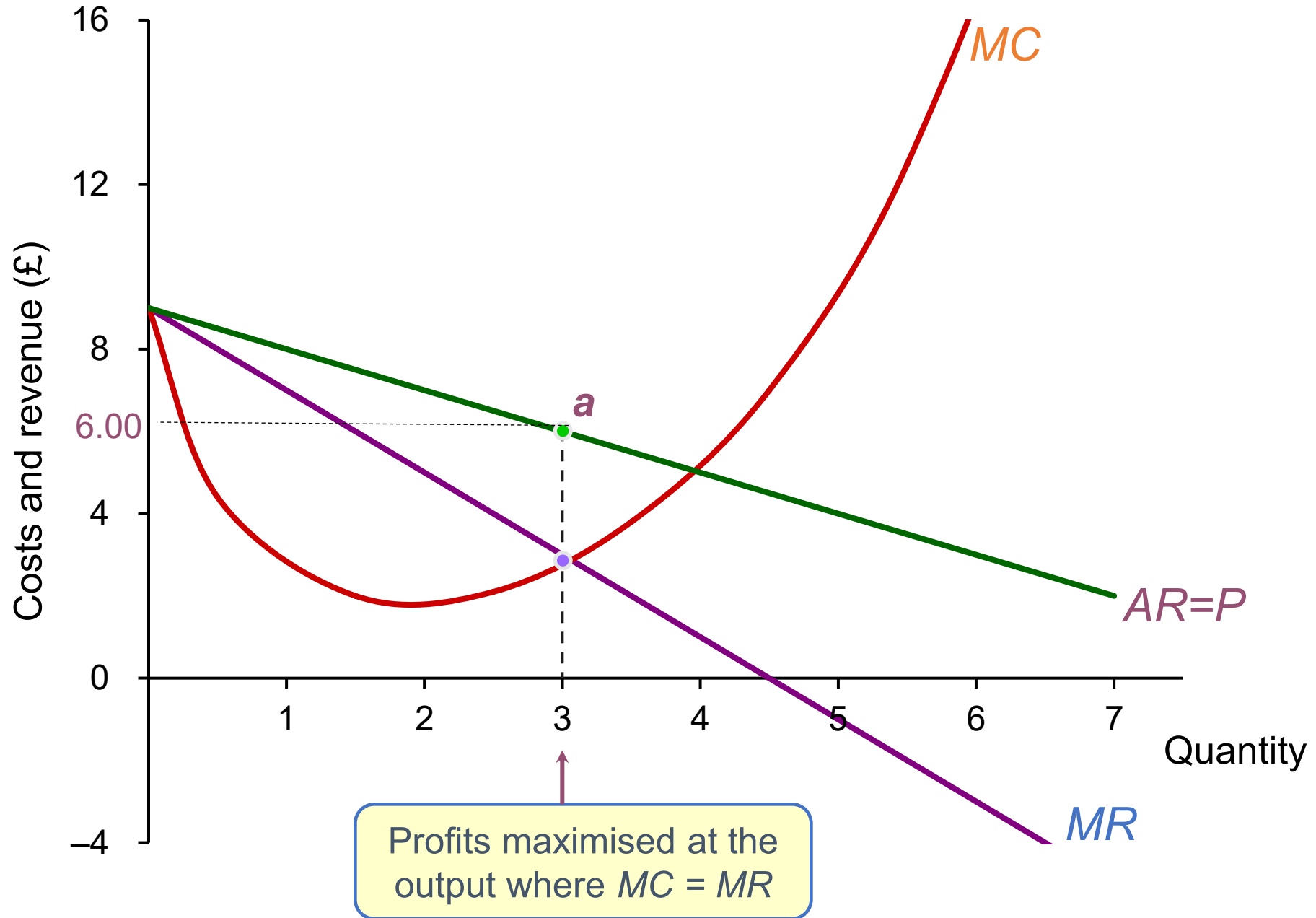
- Total Profit: Multiply Average Profit by the total Quantity (Q).
- Example:

$$(\pounds 6.00 - \pounds 4.50) \times 3 \text{ units} = \pounds 4.50 \text{ Total Profit}$$

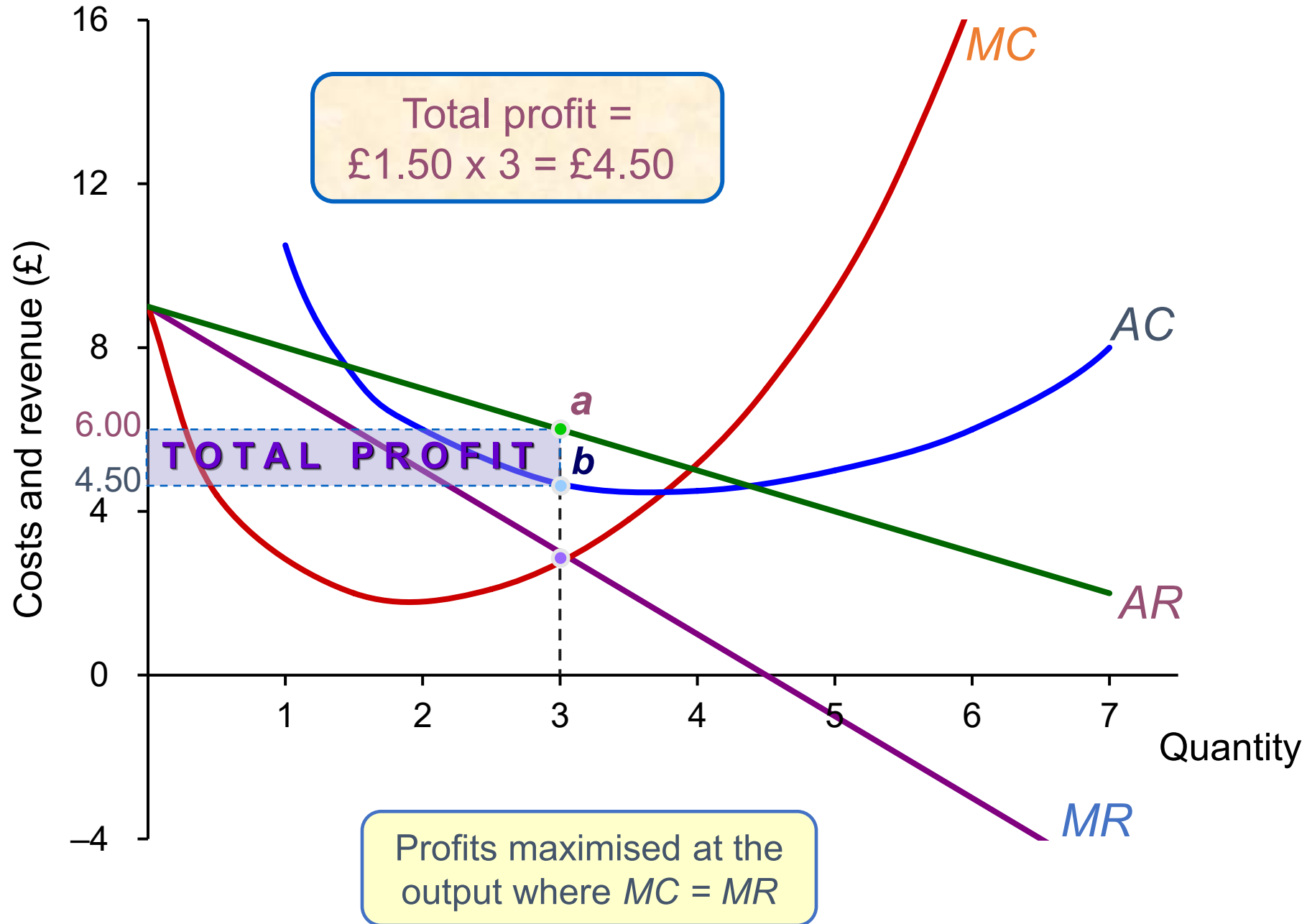
Measuring the maximum profit using average curves



Measuring the maximum profit using average curves



Measuring the maximum profit using average curves



Normal vs. Supernormal Profit

In economics, Profit is defined differently than in accounting:

- Normal Profit: This is the opportunity cost of the owners' time and capital. It is the minimum return needed to keep the business running.
 - Crucial Point: Normal profit is already included in the Cost curves (AC and TC).
- Supernormal Profit (Economic Profit): Any profit made above normal profit.
 - Occurs when $AR > AC$.
 - This is the shaded area on a graph representing pure economic gain.

Questions